

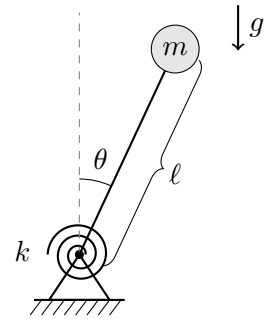
16.07 Dynamics — Quiz 1: Spring-Loaded Inverted Pendulum

Name: _____

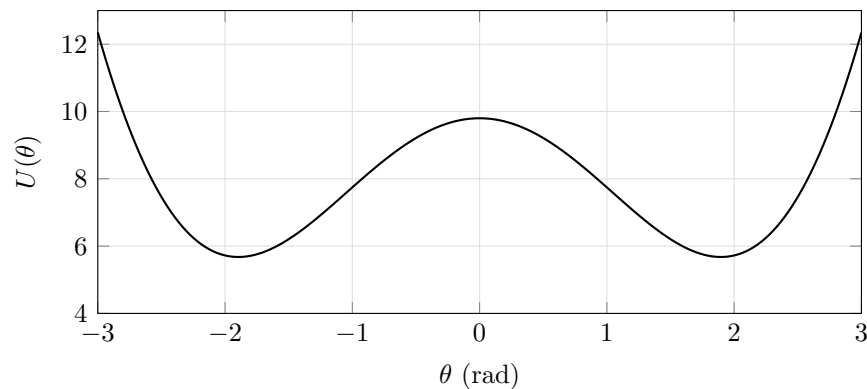
1. The nonlinear equation of motion for the system shown is

$$\ddot{\theta} = \frac{g}{\ell} \sin \theta - \frac{k}{m\ell^2} \theta.$$

Find the critical stiffness k_{cr} , in terms of m , g , and ℓ , at which the upright equilibrium transitions from stable to unstable.



2. The plot below shows $U(\theta)$ for $k = 0.5 k_{cr}$, with $m = \ell = 1$ and $g = 9.8$. Mark every stable equilibrium with a circle ($\circ$) and every unstable equilibrium with an $\times$.



3. In general, knowing $\vec{r}$ and $\vec{\tau}$, why can't $\vec{\tau} = \vec{r} \times \vec{F}$ be solved uniquely for $\vec{F}$? Why is it possible to find the spring force this way in this problem?